Future smartphone could be powered by bacteria

A team of scientists from Oxford University has showed how the natural movement of bacteria could be harnessed to assemble and power microscopic wind farms or other man-made micro-machines such as smartphone components. The team has used computer simulations to demonstrate that the chaotic swarming effect of dense active matter such as bacteria can be organised to turn cylindrical rotors and provide a steady power source.

The technique could be used to generate energy on a tiny scale and, if used widely enough, may make a big difference and help tackle the growing energy crisis (Image courtesy: www.dailymail.co.uk)

Researchers say that these biologically-driven power plants could some day be the microscopic engines for tiny, man-made devices that are self-assembled and self-powered—everything from optical switches to smartphone microphones.

Dense bacterial suspensions are the quintessential example of active fluids that flow spontaneously. While swimming bacteria are capable of swarming and driving disorganised living flows, these are normally too disordered to extract any useful power from.

Wearables can tell if you are sick before you can

New research from Stanford shows that fitness monitors and other wearable biosensors can tell when your heart rate, skin temperature and other measures are abnormal, suggesting possible illness.

Wearable sensors that monitor heart rate, activity, skin temperature and other variables can reveal a lot about what is going on inside you, including the onset of infection, inflammation and even insulin resistance, according to a study by researchers at Stanford University School of Medicine.

Geneticist Michael Snyder was wearing seven biosensors collecting data about his health when he noticed changes in his heart rate and oxygen level during a flight (Image courtesy: www.geekwire.com)

An important component of the ongoing study is to establish a range of normal, or baseline values for each person in the study and when they are ill. "We want to study people at an individual level," said Michael Snyder, PhD, professor and chair of genetics.

Lie-detecting kiosks are here

The Automated Virtual Agent for Truth Assessments in Real Time (AVATAR) is currently being tested in conjunction with Canadian Border Services Agency to help border security agents determine whether travellers coming into Canada may have undisclosed motives for entering the country.

AVATAR can detect changes in physiology and behavior during interviews with travellers (Image courtesy: San Diego State University)

According to San Diego State University management information systems professor, Aaron Elkins, "AVATAR is a kiosk, much like an airport check-in or grocery store self-checkout kiosk. However, it has a face on the screen that asks travellers questions, and can detect changes in physiology and behaviour during the interview. The system can detect changes in the eyes, voice, gestures and posture to determine potential risk. It can even tell when they are curling their toes."

Once the kiosk detects deception, it would flag those passengers for further scrutiny by human agents.

ICAROS is the fitness machine that simulates flying

ICAROS is a full-body system with a gyroscopic design, which aims to make working out more enjoyable. In this setup you have to use your core strength to control the movements of the machine and in the game itself. You have to lie in a plank-like position the entire time while flying through an array of virtual settings. You can choose games depending on your physical abilities, and work up to more difficult settings to increase the benefits of the exercise.

Icaros active VR gives you a full-body workout as you swivel to control flight (Image courtesy: www.icaros.net)